



Tito Nicola Drugman

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Milan, Italy (Home)

ABOUT MYSELF

M.Sc. Computer Science & Engineering student (AI track) at Politecnico di Milano with hands-on experience deploying open-source AI on local Linux infrastructure under strict data-sovereignty and hardware constraints - from Transformers running locally on STM32 NPUs to Retrieval-Augmented Generation pipelines over 7,600+ document knowledge bases and Vision-Language-Model document/page classification workflows, all on local hardware, minimizing third-party API dependencies. Focused on reproducible AI systems and on-premises deployment of open-source models, with attention to secure execution, confidentiality and systematic documentation of experiments and configurations. Daily Ubuntu Linux user; experienced communicating technical content to mixed audiences through multi-year certification-trainer roles.

EDUCATION & TRAINING

Master of Science in Computer Science & Engineering (Artificial Intelligence)

Politecnico di Milano [01/09/2024 - Current]

City: Milan | **Country:** Italy

Level in EQF: 7

Machine Learning, Deep Learning, Natural Language Processing, Large Language Models, Neural Networks, Distributed Systems, Software Engineering

Exchange Student – Computer Science & Engineering

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) [01/09/2025 - 01/01/2026]

City: Shenzhen | **Country:** China

Level in EQF: 7

AI research, advanced coursework in machine learning and engineering

Bachelor of Science in Artificial Intelligence

Università di Pavia / Università degli Studi di Milano / Università degli Studi di Milano-Bicocca [01/09/2021 - 01/07/2024]

City: Milan | **Country:** Italy

Level in EQF: 6

Inter-university bachelor's program. Artificial Intelligence, Machine Learning, Statistics, Algorithms, Programming, Databases

WORK EXPERIENCE

STMicroelectronics - Italy, Agrate Brianza (Milan)

AI Engineer Intern - Bachelor's Thesis

[01/03/2024 - 31/07/2024]

- Designed and implemented an automated hyperparameter optimisation framework (Python, extending the Keras Tuner library) for neural networks deployed on microcontroller-class hardware, targeting an industrial audio anomaly-detection use case for predictive maintenance of machinery.
- Engineered the search procedure to evaluate candidate architectures jointly against accuracy and strict embedded constraints - ROM footprint, RAM usage and Multiply-Accumulate operations (MACCs) - producing only deployable models.
- Validated the framework's ROM/RAM/MACC estimators against STM32Cube.AI ground-truth measurements across three MLPerf Tiny reference models - anomaly detection, image classification and keyword spotting.
- Demonstrated the end-to-end constraint-aware search on two MLPerf Tiny tasks - industrial audio anomaly detection (predictive maintenance) and keyword spotting - producing smaller, constraint-compliant models with performance comparable to the larger MLPerf

baselines.

- Diagnosed and engineered a workaround for an upstream bug in the open-source Keras Tuner library that kept long unattended remote tuning runs alive under hardware-constraint filtering.
- Operated within STMicroelectronics' confidential corporate environment: remote access to internal Linux workstations through company VPN and standard access-control protocols, including long unattended tuning runs that respected institutional security boundaries.
- Maintained end-to-end reproducibility through version-controlled Conda environments, systematic experiment logs and documented configurations later compiled directly into the bachelor's thesis.

Fanuc - Sanoma / Comau - Pearson

Industrial Robotics Trainer & Certification Assessor

[01/10/2022 - Current]

- Comau – Pearson (2022–2023)
- Fanuc – Sanoma (2023–present)
- Deliver technical training and administer official industry certifications on industrial communication protocols, robot programming and Industry 4.0 to mixed audiences of engineers, educators and students across Northern Italy.
- Produce structured course materials, assessment criteria and lead "Train the Trainer" sessions - translating dense technical content into reproducible curricula.

Leonardo da Vinci National Museum of Science and Technology - Italy, Milan

Science Museum Guide

[01/03/2023 - Current]

- Communicate complex technical and historical concepts to diverse public audiences across aviation, transport and historical technology exhibitions.

PROJECTS

[01/09/2025 - 28/02/2026]

Retrieval-Augmented Generation (RAG) for LLM-Based Code Generation in Large Codebases

- Designed a reproducible document ingestion, chunking and indexing pipeline over a 7,600+ document knowledge base - corpus-agnostic by design and directly transferable to legal, audit and technical-documentation domains.
- Engineered and benchmarked three single-hop retrievers (lexical BM25, semantic dense via CodeBERT 768-d embeddings and hybrid via Reciprocal Rank Fusion) and two LLM-driven information-extraction pipelines (query decomposition and iterative refinement) for complex multi-hop queries.
- Deployed the end-to-end RAG system fully on-premises on local Linux hardware (2xRTX 5060 Ti) with zero external API calls - used Hugging Face Transformers and 4-bit NF4 quantization (bitsandbytes) to fit CodeLlama-7B-Instruct in available VRAM, ensuring data sovereignty and confidentiality of the indexed corpus.
- Result: semantic retrieval delivered a +38% relative improvement over the no-retrieval baseline (0.2567 vs. 0.1859 CodeBLEU); multi-hop decomposition reached 0.2505.
- Production engineering: deterministic generation, resume-safe execution, OOM handling, version-controlled configurations, full end-to-end reproducibility.

Link: <https://github.com/TitoNicolaDrugman/RAG-Code-Generation>

[01/09/2025 - 01/01/2026]

Replicating and Enhancing WebVoyager - End-to-End AI Web Agents Powered by Locally-Deployed Open-Source Large Multimodal Models

- Replaced proprietary GPT-4o with locally-hosted open-source alternatives: deployed Qwen3-VL:8b and Qwen3-VL:30b via Ollama on local hardware; integrated DeepSeek-v3 (685B MoE) via OpenRouter as a third comparison point.
- Built a fully-automated document/page classification pipeline using locally-hosted Qwen3-VL:30b to classify webpage screenshots into 12 semantic categories - produced a 1,101-image labelled dataset with zero manual annotation.
- Fine-tuned a ResNet-18 CNN for real-time page-state classification, integrated as a visual prior in the agent loop; reduced reasoning latency by 17%.
- Designed a novel grayscale + colored Set-of-Mark visual grounding technique (52.4% vs. 51.1% baseline); replicated GPT-4o reference (62.7% task success, surpassing the paper's 59.1%).
- Stack: Python, Selenium, Ollama, PyTorch, Hugging Face, OpenRouter.

Link: <https://github.com/TitoNicolaDrugman/LLMsWebAgent>

[01/03/2025 - 30/09/2025]

On-Device Transformer Deployment on the NPU of an STM32N6 - Full-Stack Edge AI Engineering Under Strict Hardware Constraints

- Designed, trained and INT8-quantized a custom 4.56 M-parameter encoder-only Transformer from scratch for autoregressive text generation - the same architectural and compression principles that govern local LLM deployment on CPU/GPU.
- Applied per-channel symmetric INT8 quantization of both weights and activations using a held-out calibration set - a methodology directly transferable to compressing larger LLMs for on-premises inference under VRAM or memory constraints.
- Engineered the full end-to-end pipeline: model design (PyTorch), training on ~1.4 M words from Project Gutenberg, quantization calibration, C-code generation via STM32CubeIDE / X-Cube-AI, NPU configuration, and on-device validation.
- On-board results: 73.14 ms/token latency, 13.67 tokens/s throughput, with 257/407 ops accelerated on the NPU; produced a quantitative CNN-vs-Transformer comparison under identical hardware constraints.

Link: <https://github.com/TitoNicolaDrugman/Transformer-NPU-STM32N6>

[01/10/2025 - 30/11/2025]

GenHack 2025 - Correcting Urban Heat Island Bias in ERA5 Climate Data via Multi-Source Geospatial Machine Learning (École Polytechnique x BNP Paribas x Kayrros)

- Three-person team hackathon: built an ML pipeline to correct systematic Urban Heat Island bias in ERA5-Land coarse-resolution (9 km) climate reanalysis data, integrating four heterogeneous data sources (ERA5-Land, Sentinel-2 NDVI, ECA&D ground stations and JRC Global Surface Water) into a unified 3M+ row training dataset across IT/ES/FR/PT.
- Trained and compared Random Forest vs. HistGradientBoosting regressors with a station-based split to prevent geographic leakage; engineered a land-cover segmentation step using Sentinel-2 NDVI fused with JRC water data to resolve urban/water spectral ambiguity.
- Result: reduced ERA5 raw error from 3.01°C to 1.94°C RMSE (−35.5%), MAE by −45%, and corrected systematic urban bias by −93% on hold-out test stations. Stack: Python, scikit-learn, GeoPandas, Xarray, Dask.

Link: <https://github.com/TitoNicolaDrugman/GENHACK>

SKILLS

Local AI & On-Prem Deployment

Ollama | llama.cpp | Hugging Face Transformers (local inference) | bitsandbytes (4-bit NF4 quantization) | local Vision-Language Model deployment (Qwen3-VL via Ollama) | OpenRouter | API-free pipelines

Systems, Infrastructure & Containers

Ubuntu Linux (daily driver, dual-boot) | SSH | Bash | Docker, Podman (familiar with rootless containers) | VPN-based remote workstation access

LLMs, RAG & Information Extraction

RAG (lexical BM25, semantic CodeBERT embeddings, hybrid via Reciprocal Rank Fusion) | Information Extraction (LLM-driven query decomposition, iterative refinement, multi-hop retrieval) | document ingestion, chunking & indexing pipelines | prompt engineering | Transformer architecture | LLMs (CodeLlama, Qwen3-VL, DeepSeek-v3, Llama-3)

AI & Machine Learning

PyTorch | TensorFlow | Keras | Hugging Face Transformers | scikit-learn | Deep Learning | Neural Networks | quantization-aware deployment | Computer Vision (image classification, semantic segmentation)

Programming & Development

Python | Git | GitHub | SQL | LaTeX | C | Java

Data & Tools

NumPy | Pandas | Matplotlib | MongoDB | SQL databases | Selenium | REST APIs | Hugging Face Hub | GeoPandas | Xarray | Anaconda | Jupyter | YAML | JSON

Communication & Documentation

Technical writing | Reproducible documentation (Conda env specs, version-controlled configs) | Public speaking | Cross-audience technical explanation

LANGUAGE SKILLS

Mother tongue(s): Italian

English

LISTENING: C2 READING: C1 WRITING: C1

SPOKEN PRODUCTION: C1

SPOKEN INTERACTION: C1